

PC-05 · UV FILTERS · PERSONAL CARE

HS-PBSA

Phenylbenzimidazole Sulfonic Acid

INCI	CAS	EC	FAMILY	SUPPLIED AS
Phenylbenzimidazole Sulfonic Acid	27503-81-7	248-502-0	PC-05	Per filter

01 POSITIONING

Nine UVB molecules in the index — the raw material of every SPF architecture the market asks for.

02 HOW IT WORKS

- 01 Erythema-band absorption

Chromophores tuned to the 290–320 nm range that drives sunburn and label SPF.
- 02 Liquid + crystal synergy

Liquid filters (OMC, OCT class) dissolve crystalline boosters (EHT class), raising SPF non-linearly.
- 03 Texture engineering

Filter blend ratios set the sensory profile as much as the SPF number.

03 APPLICATIONS

- Body sunscreens SPF 30–90
- Sport and water-resistant lines
- Lip and stick SPF formats
- Hair UV protection

04 FORMULATION GUIDE

USE LEVEL	Per regional annexes
PH & CONDITIONS	Respect regional ceilings; avobenzene pairing requires stabilization strategy
— Hot-dissolve crystalline boosters in the liquid filters plus ester solvent; verify no recrystallization at 4 °C.	
— Water-resistance comes from the film former as much as the filter — engineer both.	

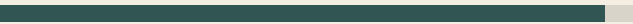
05 TYPICAL SPECIFICATION

Appearance	per filter: clear liquid / powder
Assay	≥ 98%
Peak absorbance	290–320 nm
Regulatory listing	per region, per TDS

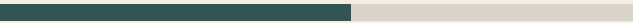
Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

SPF units per % filter (blended, class data)
Optimized UVB blend



Cost per SPF unit
Blend-engineered system



Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.